

NLP Project - I

Smart Email Classification and Routing System Using NLP

Problem statement

Modern Organizations receive thousands of emails ~~from~~ everyday from customers, partners, vendors and internal employees.

These emails can be related to

- (A) Technical issues
- (B) Billing problems
- (C) Refund Requests
- (D) Account management
- (E) Product queries
- (F) Complaints
- (G) Feedback
- (H) General Enquiries

The manual process Causes.

- (A) Delayed Responses
- (B) Increased workloads
- (C) Human errors.
- (D) Poor customer experience
- (E) Higher operational costs.

The goal of this project is to develop an intelligent system that automatically analyzes incoming email using NLP, identifies ~~and~~ intent and category of email, determines its priority level and routes it to the appropriate department.

Example 1

Email 1 → My payment failed but the money was deducted from my bank account.

Expected Result →

Category → Billing

Priority → High

Department → Finance team.

Proposed solution.

Step 1 (Receive customer mail)



Step 2 (Process the text using NLP techniques)



Step 3
Extract meaningful information.



Step 4: Predict email category



Step 5: Determine the priority level.



Step 6: Store results in Supabase



Step 7: Display results on React dashboard.



Step 8: Forward the email to correct department.

Project objectives.

- ① Reduce manual email sorting effort
- ② Improve response time
- ③ Increase customer satisfaction
- ④ Automatically classify emails
- ⑤ Identify urgent emails
- ⑥ Generate business insights from customer communication

NLP Features

① Feature 1: Text Preprocessing

Perform:

- ① Lowercasing ② Stopword removal ③ Tokenization ④ Lemmatization

Original:

My payment has failed repeatedly.

Processed:

Payment failed repeatedly.

② Feature 2: Email Classifiers.

③ Feature 3: Sentiment Analysis.
Detect customer sentiment

Positive → Thank you for resolving my issue quickly.

Negative → I am extremely disappointed with your service.

Neutral → Please provide product details.

Support managers can monitor customer satisfaction.

④ Priority detection. → Determine urgency.

High priority

Money deducted, Account hacked, Payment failed

Medium Priority

Unable to login, need password reset.

Low Priority → Need product information

Feature 5 (Keyword extraction)

Extract important words

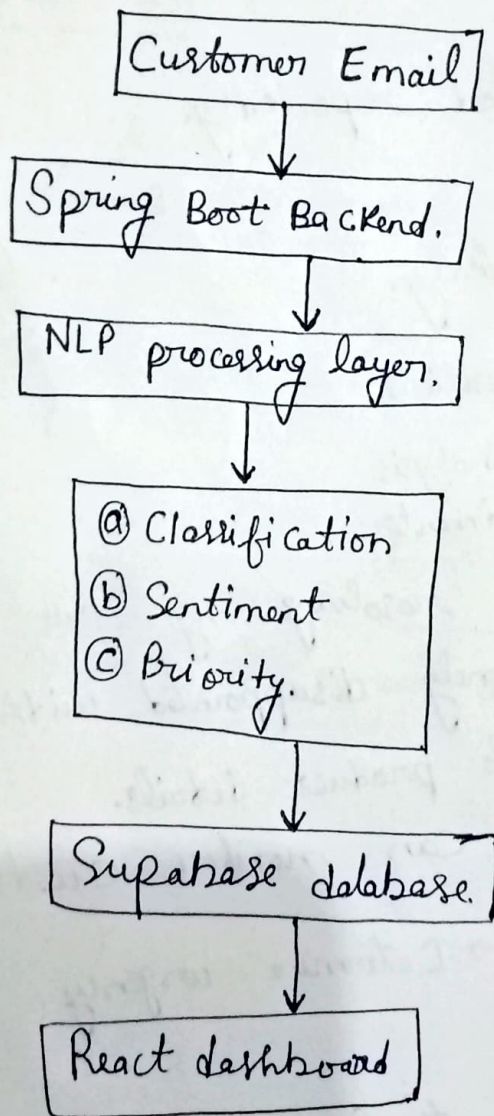
my credit card payment failed.

Output → Credit card, Payment failed.

Feature 6 (Named entity recognition)

- (a) Person names
- (b) Organizations
- (c) Dates
- (d) Amounts.
- (e) Transaction IDs

System Architecture.

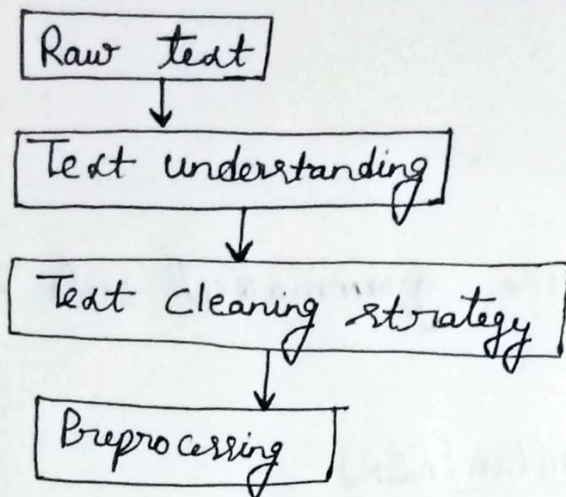


Steps to build project

Step 1 $\Rightarrow$ Download the dataset and open google Colab.

Step 2 $\Rightarrow$ Exploratory data analysis (EDA)
(Understand the data well)

Step 3 $\rightarrow$ Text quality Analysis & NLP data understanding.
Understand the raw text before preprocessing



Step 4 Text Cleaning.

This is where we prepare the raw emails for NLP preprocessing.

- (a) Convert text to lowercase
- (b) Remove URLs
- (c) Remove Email addresses.
- (d) Remove special characters
- (e) Remove extra spaces.
- (f) Keep important business information.

Step 5 $\Rightarrow$ Tokenization

Tokenization means breaking text into smaller units. Called tokens.

tokenization is the foundation of.

- | | |
|----------------------|----------------|
| (i) Stopword Removal | (iv) NER |
| (ii) Lemmatization | (v) TF-IDF |
| (iii) Pos tagging | (vi) word vec. |

Step 6 Stopword Removal.

Stopwords are words that occur very frequently but usually do not contribute much meaning to Classify.

Step 7 Lemmatization.

Lemmatization converts words to their dictionary root form (lemma).

Step 8 POS Tagging

POS tagging identifies the grammatical role of each word.

Step 9: Named Entity Recognition (NER).

NER identifies and classifies important entities in text we are going to use spacy here.

Step 10 → Keyword Extraction.

Keyword extraction identifies the most important words from a documents.

Step 11 → Feature Engineering using TF-IDF.

Machine learning models can not understand the text, they only understand numbers so we convert text into numerical vectors.

Step 12. Model Benchmarking.

We will compare multiple classification model

Step 13 model training

Select the best model.

Step 14 model persistence (Saving the model).

Step 15 $\Rightarrow$ Create prediction pipeline